# Aditya Rajesh Achary

linkedin.com/in/aditya-achary
github.com/in/aditya-achary

Email: aditya.achary0909@gmail.com
Mobile: +91-9322888601

## Projects

- **AI Adaptive Learning Platform** *2026*
  - Architected a multi-agent pipeline (LangGraph, Gemini) generating verified question variants via blind-solver validation.
  - Built an adaptive diagnosis engine classifying student misconceptions to auto-generate targeted prerequisite drills.
  - Designed a hybrid RAG tutor (FastEmbed, MongoDB) streaming Socratic hints via SSE with anti-leak guardrails.
  - Parsed 8,600+ questions from 170+ PDFs via PyMuPDF to generate syllabus primers, formula sheets, and mind maps.
- **Next-Generation Prosthetic Hand with Intelligent Control** *2025 – 2026*
  - Designed a hybrid CvT-SLA-GGCNN model that cut parameter count while preserving grasp detection accuracy, improving deployability on edge hardware
  - Built and labeled a custom multi-class grasp dataset (box, cylindrical, precision) to boost model generalization across real-world objects
  - Deployed a real-time webcam-based inference pipeline enabling adaptive, low-latency prosthetic control
- **Smart Helmet: Edge AI Multimodal Driver Assistance & Proximity Safety System** *2025*
  - Built an edge-based ADAS combining YOLOv8 object detection and Depth-Anything ViT depth estimation, achieving 30 FPS real-time inference on CUDA
  - Engineered a spatial proximity algorithm converting depth maps into tiered collision alerts (DANGER, WARNING, SAFE) for driver safety
  - Integrated GPS and alcohol-sensor telemetry over serial communication for real-time hardware-triggered safety alerts

## Experience

**Ashok Leyland** Chennai, India
*Data Analyst Intern* *Apr 2025 – May 2025*

- Engineered Python and SQL-based data pipelines, improving processing efficiency by 5% through scalable ETL workflows.
- Developed hybrid recommendation models (collaborative filtering, SVD) to enhance recommendation performance.
- Validated and debugged structured datasets to ensure reliability and consistency of model outputs.

## Skills Summary

- **Languages:** Python, TypeScript, JavaScript, SQL, R
- **Generative AI & Agents:** LangGraph, LangChain, RAG, Multi-Agent Systems, FastEmbed, Vector Databases, Gemini API, Prompt Engineering
- **Machine Learning & CV:** PyTorch, TensorFlow, OpenCV, YOLO, DepthAnything, MiDaS, Scikit-learn
- **Backend & Data Pipelines:** FastAPI, PyMuPDF, ETL Pipelines, MongoDB, REST APIs, Server-Sent Events (SSE)
- **Frontend & Tools:** React, Tailwind CSS, Git, Docker, Pytest, Power BI, MATLAB

## Education

**Amrita Vishwa Vidyapeetham** Kerala, India
*Bachelor of Technology – Electronics and Computer Engineering; CGPA: 8.14/10* *2022 – 2026*
**Campion School, Cochin** Kerala, India
Senior Secondary: 90% Secondary: 96.6% *2020 – 2022*

## Publications

- **Advanced Song Recommendation Framework (Machine Learning):** Built a hybrid recommendation engine achieving 94.2% accuracy across 10 genres.
- **IoT Real-Time Bus Information System (Computer Vision, IoT):** Developed a machine learning-based bus detection system using ESP32, YOLO, and EasyOCR for real-time tracking.

## Certifications

**Google AI Professional Certification** *Google, 2026*
**IBM GenAI Certification** *IBM, 2026*

## Leadership

**Student Social Responsibility (SSR) Project** Kerala, India
Conducted hands-on sessions on computational thinking and AI concepts for 80+ school students. *2024*